

Learning Lean 4

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Abstract

Working notes and exercises from learning the Lean 4 theorem prover, following *An Introduction to Lean 4* by Cosme Llop   and Gong. Covers basic syntax and natural-deduction proofs, paired with a runnable Lean library.

1. BASIC SYNTAX

1.a. Basic syntax

The runnable version of this chapter lives in [LearningLean/BasicSyntax.lean](#).

1.a.i. Comments:

```
-- This is a single-line comment.

/-
This is a multi-line comment.
Here is another line.
- /
```

1.a.ii. Inspecting terms and types with #check:

#check asks Lean for the type of an expression without evaluating it.

```
#check true           -- Bool
#check 42              -- Nat
#check "Hello, World!" -- String
#check Type            -- Type 1
#check Nat             -- Type
#check ['h', 'e', 'l', 'l', 'o'] -- List Char
```

1.a.iii. Printing definitions with #print:

#print shows how something is defined.

```
#print Bool
```

Type itself cannot be printed with #print, because it is a type rather than a definition.

1.a.iv. Definitions and functions:

```
def pi : Float := 3.1415926

#check pi
```

```
-- An anonymous function (lambda) taking two natural numbers.  
#check λ (a b : Nat) => a + b
```

2. NATURAL DEDUCTION

2.a. Natural deduction

The Lean proofs for this chapter live in [LearningLean/NaturalDeduction.lean](#), inside the Exercises namespace.

2.a.i. 5.5 Reduction to the absurd. $P \Rightarrow Q, \neg Q \vdash \neg P$:

This is a very useful technique. Validity of $P \Rightarrow Q, \neg Q \vdash \neg P$ is proved with:

1	$P \Rightarrow Q$	
2	$\neg Q$	
3	$\begin{array}{ l} P \end{array}$	H
4	$\begin{array}{ l} Q \end{array}$	$E \Rightarrow 1,3$
5	$\begin{array}{ l} \neg Q \end{array}$	IT 2
6	$\neg P$	$I \neg 3,4,5$

The sub-derivation (lines 3–5) discharges the hypothesis P : assuming P , [$\Rightarrow$ -elimination] on lines 1 and 3 gives Q , while line 2 reiterates $\neg Q$. The contradiction lets us introduce $\neg P$ ($\neg$ -introduction over lines 3, 4, 5).

In Lean this is theorem T55, where $\neg P$ is definitionally $P \rightarrow \text{False}$, so the whole proof is just function composition:

```
theorem T55 (h1 : P → Q) (h2 : ¬Q) : ¬P :=
  (h2 ∘ h1)
```

2.a.ii. Conjunction introduction. $P, P \Rightarrow Q \vdash P \wedge Q$:

Theorem T51 builds the pair (proof of P , proof of Q) directly, using the anonymous-constructor notation $\{_, _\}$:

```
theorem T51 (h1 : P) (h2 : P → Q) : P ∧ Q :=
  {h1, h2 h1}
```